

American Option Pricing and Markovian Projections

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1 Bachelier and Black Scholes Model

In the Bachelier model, the price process $X(t)$ satisfies

$$dX(t) = \mu dt + \sigma dW(t), \quad X(0) = x_0 \in \mathbb{R}. \quad (1)$$

Integrating from 0 to t ,

$$X(t) = x_0 + \mu t + \sigma W_t. \quad (2)$$

Because $W_t \sim \mathcal{N}(0, t)$, the conditional distribution is

$$X(t) | X(0) = x_0 \sim \mathcal{N}(x_0 + \mu t, \sigma^2 t). \quad (3)$$

The transition density is

$$\phi_{\text{Bach}}(t, y; x_0) = \frac{1}{\sqrt{2\pi\sigma^2 t}} \exp\left(-\frac{(y - x_0 - \mu t)^2}{2\sigma^2 t}\right), \quad y \in \mathbb{R}. \quad (4)$$

In the Black Schole model, the asset price $S(t)$ obeys

$$dS(t) = \mu S(t) dt + \sigma S(t) dW(t), \quad S(0) = s_0 > 0. \quad (5)$$

Dividing by $S(t)$ and integrating using the Ito formula,

$$\ln S(t) = \ln s_0 + \left(\mu - \frac{\sigma^2}{2}\right)t + \sigma W_t. \quad (6)$$

Therefore

$$\ln S(t) \mid S(0) = s_0 \sim \mathcal{N}\left(\ln s_0 + \left(\mu - \frac{\sigma^2}{2}\right)t, \sigma^2 t\right). \quad (7)$$

Let

$$\mu_\ell = \ln s_0 + \left(\mu - \frac{\sigma^2}{2}\right)t, \quad \sigma_\ell^2 = \sigma^2 t.$$

Changing variables from $\ell = \ln s$ back to $s = e^\ell$ yields the log-normal density

$$\phi_{\text{BS}}(t, s; s_0) = \frac{1}{s \sigma \sqrt{2\pi t}} \exp\left(-\frac{(\ln s - \mu_\ell)^2}{2\sigma^2 t}\right), \quad s > 0. \quad (8)$$

2 Markovian Projections

Lemma 1 (Markovian Projection, one-dimensional version). *Let X solve (254) and consider the non-Markovian process $S_1 = P_1 X$. Let*

$$\begin{aligned} d\bar{S}^{(x_0)}(t) &= \bar{a}^{(x_0)}(t, \bar{S}^{(x_0)}(t)) dt + \bar{b}^{(x_0)}(t, \bar{S}^{(x_0)}(t)) d\bar{W}(t), & t \in [0, T], \\ \bar{S}^{(x_0)}(0) &= P_1 x_0. \end{aligned} \quad (9)$$

where $\bar{W}$ is a Wiener process and the drift and the diffusion coefficients of the surrogate process $\bar{S}^{(x_0)}$ are determined as follows

$$\bar{a}^{(x_0)}(t, s) = \mathbb{E}[P_1 a(t, X(t)) \mid P_1 X(t) = s, X(0) = x_0] \quad (11)$$

$$(\bar{b}^{(x_0)}(t, s))^2 = \mathbb{E}[(P_1 b b^\top P_1^\top)(t, X(t)) \mid P_1 X(t) = s, X(0) = x_0] \quad (12)$$

Then, $S_1(t) \big|_{\{X(0)=x_0\}} = P_1 X(t) \big|_{\{X(0)=x_0\}}$ and $\bar{S}^{(x_0)}(t)$ have the same conditional distribution for all $t \in [0, T]$.

Exercise 1. Consider the system

$$dX_i(t) = rX_i(t) dt + \sigma X_i(t) dW_i(t), \quad (13)$$

with $r = 0.02$ and $\sigma = 0.05$. The Wiener process above has independent components. Compute, using discrete L^2 regression, the dynamics for the one-dimensional Markovian projection of the non-Markovian

$$S = \frac{1}{d} \sum_{i=1}^d X_i, \quad (14)$$

corresponding to $X_i(0) = 1$ and $0 \leq t \leq T = 2$. Take $d = 30$.

Solution. Markovian Projection: This is the same setting as in exercise 7.7 with the difference that we use the projection

$$P = \frac{1}{d}(1, \dots, 1) \quad (15)$$

instead of P_1 . The underlying SDE is:

$$dX_i(t) = a_i(t, X(t)) dt + \sum_{j=1}^k b_{ij}(t, X(t)) dW_j(t), \quad i = 1, \dots, d \quad (16)$$

$$X(0) = x_0 \quad (17)$$

where

$$x_0 = (1, \dots, 1)^\top \quad (18)$$

$$a_i(t, x) = rx_i \quad (19)$$

$$b_{i,j}(t, x) = \begin{cases} \sigma x_i, & \text{for } i = j \\ 0, & \text{otherwise} \end{cases} \quad (20)$$

We consider the non-Markovian process $S(t) = PX(t)$. Let

$$\begin{aligned} d\bar{S}^{(x_0)}(t) &= \bar{a}^{(x_0)}(t, \bar{S}^{(x_0)}(t)) dt + \bar{b}^{(x_0)}(t, \bar{S}^{(x_0)}(t)) d\bar{W}(t), \quad t \in [0, T] \\ \bar{S}^{(x_0)}(0) &= Px_0 \end{aligned} \quad (21)$$

where $\bar{W}$ is a Wiener process. The drift and diffusion coefficients in (21) are given by

$$\bar{a}^{x_0}(t, s) = \mathbb{E}[Pa(t, X(t)) \mid PX(t) = s, X(0) = x_0] \quad (22)$$

$$= \mathbb{E}[rPX(t) \mid PX(t) = s, X(0) = x_0] = rs \quad (23)$$

$$(\bar{b}^{x_0})^2(t, s) = \mathbb{E}[Pbb^\top P^\top(t, X(t)) \mid PX(t) = s, X(0) = x_0] \quad (24)$$

$$= \mathbb{E} \left[\frac{1}{d^2} \sum_{i=1}^d \sigma^2 X_i^2(t) \mid PX(t) = s, X(0) = x_0 \right] \quad (25)$$

Approximate $\bar{b}^{x_0}$: We recall that, for random variables X and Y , the conditioned expected value $\mathbb{E}[X \mid Y] = g(Y)$ satisfies the best mean square approximation:

$$\mathbb{E}[(X - g(Y))^2] \leq \mathbb{E}[(X - h(Y))^2] \quad (26)$$

for all functions h with $\mathbb{E}[h(Y)^2] < \infty$ where

$$g(Y) = \arg \min_{h \in V} \mathbb{E}[(X - h(Y))^2], \quad (27)$$

where $V = \{h : \mathbb{E}[h(Y)^2] < \infty\}$.

- (a) Observe that $g(y) = \mathbb{E}[X|Y = y]$ in the previous remark. This in some sense our $(\bar{b}^{x_0})^2(t, s) = \mathbb{E}[Z(t)|S(t) = s]$ but now the random variables depending on t . However, we want to approximate b at different values of s . For this we would like to approximate h as a function taking any value in s and not only $h(s)$. To do that, we find $h(Y)$ as a function derived from a regression as in (27). In other words, regression (27) will give us a function h and not only a value $h(s)$

Using (27), we have two options to derive $(\bar{b}^{x_0})^2$:

1. For $0 < t < T$, we have

$$(\bar{b}^{x_0})^2(t, \cdot) \approx \arg \min_{h \in V} \mathbb{E} \left[((P_1 b b^\top P_1^\top)(t, X(t)) - h(PX(t)))^2 \mid X(0) = x_0 \right] \quad (28)$$

$$= \arg \min_{h \in V} \mathbb{E} \left[\left(\frac{1}{d^2} \sum_{i=1}^d \sigma^2 X_i^2(t) - h(PX(t)) \right)^2 \mid X(0) = x_0 \right] \quad (29)$$

where V is an *ansatz* function space to model space, for example $V = \{h : h(x) = c_0 + c_1 x\}$. We can approximate (29) by a MC estimator:

$$(\bar{b}^{x_0})^2(t, \cdot) \approx \arg \min_{h \in V} \frac{1}{M} \sum_{m=1}^M \left(\frac{1}{d^2} \sum_{i=1}^d \sigma^2 (X_i^m(t))^2 - h(PX^m(t)) \right)^2 \quad (30)$$

where $X^m(t)$ are independent realizations of (13).

- (b) From approach 1, we see that $(\bar{b}^{x_0})^2(t, \cdot) = \varepsilon(t)$ is a minimizer nonnegative value ($\mathbb{E}[\text{something}^2]$). Thus the sum of all of these kinds of errors $\varepsilon(t)$ is also minimum, hence integrating in time allows us to estimate the coefficient $(\bar{b}^{x_0})^2$ in both, time and space simultaneously

2. Alternatively, we model the coefficients in space and domain:

$$(\bar{b}^{x_0})^2(\cdot, \cdot) \approx \arg \min_{h \in V} \int_0^T \mathbb{E} \left[((P_1 b b^\top P_1^\top)(t, X(t)) - h(t, PX(t)))^2 \mid X(0) = x_0 \right] dt \quad (31)$$

$$= \arg \min_{h \in V} \int_0^T \mathbb{E} \left[\left(\frac{1}{d^2} \sum_{i=1}^d \sigma^2 X_i^2(t) - h(t, PX(t)) \right)^2 \mid X(0) = x_0 \right] dt \quad (32)$$

where V is an *ansatz* function space to model space and time, for example $V = \{h : h(t, x) = c_0 + c_1 t + c_2 x\}$. We can approximate (32) by discretizing the time and applying a MC estimator for the expected value. Therefore, we consider a grid $0 = t_0 < t_1 < \dots < t_N = T$:

$$(\bar{b}^{x_0})^2(\cdot, \cdot) \approx \arg \min_{h \in V} \frac{1}{M} \sum_{m=1}^M \frac{1}{N} \sum_{n=1}^N \left(\frac{1}{d^2} \sum_{i=1}^d \sigma^2 X_i^m(t_n)^2 - h(t_n, PX^m(t_n)) \right)^2 \quad (33)$$

where $X^m(t)$ are independent realizations of (13). This is a regression problem.

For our *simulations* we consider the second approach. We assume to have the monomial basis $(\phi_0, \phi_1, \dots)$ with $\phi_i(x) = x^i$. To build a basis of polynomials on $\mathbb{R}^2$, we use $(\phi_{(i_1, i_2)})_{(i_1, i_2) \in \Lambda}$ for a two dimensional index set Λ and

$$\phi_{(i_1, i_2)} : \mathbb{R}^2 \rightarrow \mathbb{R}, \quad (34)$$

with $\phi_{(i_1, i_2)}(t, x) = \phi_{i_1}(t) \cdot \phi_{i_2}(x) = t^{i_1} x^{i_2}$. The ansatz function is based on this basis

$$\hat{h}(t, s) = \sum_{(i_1, i_2) \in \Lambda} c_{i_1, i_2} \phi_{(i_1, i_2)}(t, s), \quad (35)$$

where $\{c_{i_1, i_2}\}_{(i_1, i_2) \in \Lambda}$ are the unknown parameters.

Derivation of the normal equation: For a given indexing $k = (n, m)$, the corresponding design matrix is

$$D_{(m, n), p} = \phi_p(t_n, PX^m(t_n)), \quad \text{for } m = 1, \dots, M, \quad n = 0, \dots, N-1, \quad p \in \Lambda \quad (36)$$

such that $\{D_{k, p}\}_{k=1, \dots, NM, p \in \Lambda} \in \mathbb{R}^{NM \times |\Lambda|}$. Further, we set

$$\psi_k = \frac{1}{d^2} \sum_{i=1}^d \sigma^2 X_i^m(t_n)^2 \quad (\psi \in \mathbb{R}^{NM}). \quad (37)$$

The optimal coefficients are given by

$$\mathbf{c}^* = \arg \min_{\{c_{i_1, i_2}\}_{(i_1, i_2) \in \Lambda}} \frac{1}{MN} \sum_{m=1}^M \sum_{n=0}^{N-1} \left(\frac{1}{d^2} \sum_{i=1}^d \sigma^2 X_i^m(t_n)^2 - \sum_{(i_1, i_2) \in \Lambda} c_{i_1, i_2} \phi_{(i_1, i_2)}(t_n, PX^m(t_n)) \right)^2 \quad (38)$$

$$= \arg \min_{\mathbf{c} \in \Lambda} (\psi - D\mathbf{c})^\top (\psi - D\mathbf{c}) \quad (39)$$

$$= \arg \min_{\mathbf{c} \in \Lambda} \underbrace{\psi^\top \psi - 2\mathbf{c}^\top D^\top \psi + \mathbf{c}^\top D^\top D \mathbf{c}}_{=: J(\mathbf{c})} \quad (40)$$

We minimize $J(\mathbf{c})$ with respect to $\mathbf{c}$

$$\frac{\partial J(\mathbf{c})}{\partial \mathbf{c}} = -2D^\top \psi + 2D^\top D \mathbf{c} \stackrel{!}{=} 0 \quad (41)$$

to obtain the normal equations

$$(D^\top D) \mathbf{c} = D^\top \psi \quad (42)$$

Numerical experiment:

1. Generate and save $M = 100$ realizations of $\{X(t_n) : n = 0, \dots, N\}$ using forward Euler.
2. Derive the design matrix D and the right hand side ψ for a basis with index set

$$\Lambda = \{(i_1, i_2) : i_1, i_2 \in (0, 1, 2, 3)\}.$$

3. Derive the coefficients $\mathbf{c}$ by solving the normal equations. This gives an approximation of the projected diffusion coefficients $(\bar{b}^{x_0})^2(\cdot, \cdot)$.
4. We test the Markovian Projection by simulating $M_{\text{test}} = 10^4$ Markovian projection paths based on $\bar{b}^{x_0}$. We also simulate $M_{\text{test}} = 10^4$ forward Euler paths and apply $P(\cdot)$ at final time. The empirical distribution at final time is compared in Figure 1.

Note that for a complete analysis, the parameter set Λ and the parameter M would require some investigation.

Figure 1 shows that accordingly to theory, the conditional distributions of $S(t)|_{\{X(0)=x_0\}} = PX(t)|_{\{X(0)=x_0\}}$ and $\bar{S}^{(x_0)}(t)$ are similar. The above choice of M and Λ seem to be sufficient to derive decent fit of the exact distribution with Markovian projection.

Remark 1. The regression becomes more stable with an orthogonal polynomial basis wrt to the density of $PX(t)$. Since the exact density is unknown, one can use an empirical density to derive an inner product and an orthogonal basis wrt to this empirical inner product (Gram Schmidt).

□

Example 1 (3-to-1 dimensional Black-Scholes model.). As the first test parametrization of the Black-Scholes model we consider the case of a correlated 3-dimensional Black-Scholes model. We decompose the volatility function into the individual volatilities, $\boldsymbol{\sigma}$, and the correlation structure of asset returns. We denote with $\mathbf{G}$ the Cholesky decomposition of the correlation matrix of the log-returns

$$\Sigma_{ij} = \sigma_i G_{ij}. \quad (43)$$

We set the numerical parameters of our test case to

$$\begin{aligned} r &= 0.05, \\ \boldsymbol{\sigma} &= (0.2, 0.15, 0.1)^T, \\ \mathbf{G}\mathbf{G}^T &= \begin{pmatrix} 1 & 0.8 & 0.3 \\ 0.8 & 1 & 0.1 \\ 0.3 & 0.1 & 1 \end{pmatrix}, \end{aligned}$$

and a portfolio of equally weighted assets

$$\mathbf{P}_1 = [1, 1, 1], \quad (44)$$

as a representative test case of three moderately correlated assets in a high short rate environment. The projected local volatility features noticeable skew, as shown in Figures ?? and ??.

3 Section 3 in Implied Stopping rules for American Option Pricing

3.1 Evaluation of the local volatility $(\bar{b}^{x_0})^2$

1. Choose a time grid: $0 < t_1 < \dots < t_N = T$ (typically the same grid that will be used for the backward finite difference solver). For every t_n simulate $M \approx 100$ paths of the d -dimensional model. From these paths set

$$S^-(t_n) = \min_i \mathbf{P}_1 X(t_n, \omega_i) \quad 0 \leq n \leq N_t, \quad 1 \leq i \leq M, \quad (45)$$

$$S^+(t_n) = \max_i \mathbf{P}_1 X(t_n, \omega_i) \quad 0 \leq n \leq N_t, \quad 1 \leq i \leq M. \quad (46)$$

2. On the interval $[S^-(t_n), S^+(t_n)]$ they pick “a few dozen points equispaced in s ”. At each of those (t_n, s_j) nodes they apply the Laplace approximation (find the mode z^* by Newton iteration and plug formula (41)) to obtain the projected local variance $\tilde{b}^2(t_n, s_j)$.
3. For every time level t_n , they fit a **third-degree polynomial** in s through the Laplace-evaluated points (blue crosses in Fig. 1a). The resulting red curve gives $\tilde{b}(t_n, s)$ for all s in the interval — and, by mild extrapolation, slightly outside it.

3.2 Numerical value function

The resulting HJB equation describes the cost-to-go function $\bar{u}_A$ of an American option written on the portfolio that has the projected dynamics:

$$\begin{aligned} -\partial_t \bar{u}_A(t, s) &= (\bar{\mathcal{L}} \bar{u}_A)(t, s) \mathbf{1}_{\max((\bar{\mathcal{L}} \bar{u}_A)(t, s), \bar{u}_A(t, s) - g(s)) > 0} \\ &= (\bar{\mathcal{H}} \bar{u}_A)(t, s), \quad (t, s) \in [0, T] \times \bar{D}, \\ \bar{u}_A(T, \cdot) &= g(\cdot). \end{aligned} \quad (47)$$

where

$$(\bar{\mathcal{L}} \bar{u}_A)(t, s) = -r \bar{u}_A(t, s) + \bar{a}^{(x_0)}(t, s) \partial_s \bar{u}_A(t, s) + \frac{(\bar{b}^{(x_0)})^2(t, s)}{2} \partial_{ss}^2 \bar{u}_A(t, s), \quad t \in [0, T], \quad s \in \bar{D}, \quad (48)$$

We set to define a finite-difference approximation $\bar{\bar{u}}_A$ of the value function $\bar{u}_A$ that solves (47), based on the finite differences operator

$$\begin{aligned} (\bar{\bar{\mathcal{L}}} \bar{\bar{u}})(t, s_n) &= \left(\frac{\bar{b}^2(t, s_n)}{2\Delta s^2} + \frac{r s_n}{2\Delta s} \right) \bar{\bar{u}}(t, s_{n-1}) - \left(r + \frac{\bar{b}^2(t, s_n)}{\Delta s^2} \right) \bar{\bar{u}}(t, s_n) \\ &\quad + \left(\frac{\bar{b}^2(t, s_n)}{2\Delta s^2} - \frac{r s_n}{2\Delta s} \right) \bar{\bar{u}}(t, s_{n+1}), \quad 1 < n < N_s, \end{aligned} \quad (49)$$

We use a stable backward Euler scheme,

$$\bar{u}_A(t_{n-1}^+, s_m) = \bar{u}_A(t_n, s_m) + (\overline{\mathcal{L}u})(t_{n-1}^+, s_m) \Delta t_n, \quad 1 \leq n \leq N_t, \quad 1 \leq m \leq N_s, \quad (50)$$

$$\bar{u}_A(t_{n-1}, s_m) = \max(\bar{u}_A(t_{n-1}^+, s_m), g(s_m)), \quad 1 \leq n \leq N_t, \quad 1 \leq m \leq N_s, \quad (51)$$

$$\bar{u}_A(t_N, s_m) = g(s_m), \quad 1 \leq m \leq N_s, \quad (52)$$

with the artificial Dirichlet-type boundary condition (see Remark 2.2) imposed by the payoff

$$\bar{u}_A(t_n, s_1) = g(s_1), \quad (53)$$

$$\bar{u}_A(t_n, s_{N_s}) = g(s_{N_s}), \quad (54)$$

and a homogeneously spaced, time-independent, mesh $s_m = m\Delta s$. The choice of the boundary condition has been discussed in the variational setting by (?). The upper bound s_{N_s} has to be chosen based on the magnitude of the drift and the volatility for the problem at hand.

4 Multilevel Weighted Least Square Polynomial Approximation

4.1 Orthogonal Polynomials

We know that the polynomials are dense in the Hilbert space $L^2([a, b])$. So, we would like to find an orthonormal basis for $L^2([a, b])$ which consists of polynomials. In addition to finite intervals, we would also like to allow semi-infinite or infinite intervals, so we would like to allow $a = -\infty$ or $b = \infty$. For such a semi-infinite or infinite interval, polynomials do not have a finite norm relative to the standard scalar product on $L^2([a, b])$. To rectify this, we have to consider the Hilbert spaces $L_w^2([a, b])$ with an integrable weight function w and a scalar product defined by

$$\langle f, g \rangle = \int_a^b w(x) f(x) g(x) dx, \quad (55)$$

and choose w appropriately. Thinking about the types of intervals, that is, finite intervals $[a, b]$, semi-infinite intervals $[a, \infty]$ or an infinite interval $[-\infty, \infty]$ and corresponding suitable weight functions w will lead to a classification of different types of orthogonal polynomials which we discuss in the following subsection.

4.2 Legendre Polynomials

The Legendre polynomials are defined on the finite interval $[a, b] = [-1, 1]$, the weight function is simply $w(x) = 1$, so that the relevant Hilbert space is $L^2([-1, 1])$. They are conventionally denoted by P_n . Their Rodriguez formula reads

$$P_n(x) = \frac{1}{2^n n!} \frac{d^n}{dx^n} (x^2 - 1)^n \quad (56)$$

They are symmetric for n even and anti-symmetric for n odd, so $P_n(x) = (-1)^n P_n(-x)$. Since $(x^2 - 1)^n = x^{2n} + nx^{2n-2} + \dots$ we can easily read off the coefficients k_n and k'_n of the monomials x^n and x^{n-1} in P_n as

$$k_n = \frac{(2n)!}{2^n (n!)^2}, \quad k'_n = 0. \quad (57)$$

For the normalisation of P_n we find from the Rodriguez formula

$$h_n = \|P_n\|^2 = \int_{-1}^1 dx P(x)^2 = \frac{1}{2^n n!} \int_{-1}^1 dx P_n(x) \frac{d^n}{dx^n} (x^2 - 1)^n \quad (58)$$

$$= \frac{1}{2^n n!} \int_{-1}^1 dx \frac{d^n P_n}{dx^n}(x) (1 - x^2)^n = \frac{k_n}{2^n} \int_{-1}^1 dx (1 - x^2)^n = \frac{2}{2n + 1}, \quad (59)$$

where we have integrated by parts n times. This means the associated basis of ortho-normal polynomials on $L^2([-1, 1])$ is

$$\hat{P}_n = \sqrt{\frac{2n + 1}{2}} P_n, \quad (60)$$

and functions $f \in L^2([-1, 1])$ can be expanded as

$$f = \sum_{n=0}^{\infty} \langle \hat{P}_n, f \rangle \hat{P}_n, \quad (61)$$

or, more explicitly, shifting the normalisation factors into the integral, as

$$f(x) = \sum_{n=0}^{\infty} a_n P_n(x), \quad a_n = \frac{2n + 1}{2} \int_{-1}^1 dx P_n(x) f(x). \quad (62)$$

The recursion formula is

$$(n + 1)P_{n+1}(x) = (2n + 1)xP_n(x) - nP_{n-1}(x) \quad (63)$$

for the Legendre polynomials. From the Rodriguez formula we can easily compute the first few Legendre polynomials:

$$P_0(x) = 1, \quad P_1(x) = x, \quad P_2(x) = \frac{1}{2}(3x^2 - 1), \quad P_3(x) = \frac{1}{2}(5x^3 - 3x), \quad P_4(x) = \frac{1}{8}(35x^4 - 30x^2 + 3). \quad (64)$$

On $[0, 1]$ we use the **shifted** versions:

$$S_n(x) := P_n(2x - 1), \quad x \in [0, 1]. \quad (65)$$

The shifted polynomials satisfy

$$\int_0^1 S_n(x) S_m(x) dx = \frac{\delta_{nm}}{2n + 1}. \quad (66)$$

To make them **orthonormal**, multiply by $\sqrt{2n+1}$:

$$\boxed{\hat{P}_n(x) = \sqrt{2n+1} P_n(2x-1)} \Rightarrow \int_0^1 \hat{P}_n \hat{P}_m = \delta_{nm}. \quad (67)$$

The first few polynomials are

$$\hat{P}_0(x) = 1, \quad (68)$$

$$\hat{P}_1(x) = \sqrt{3}(2x-1), \quad (69)$$

$$\hat{P}_2(x) = \sqrt{5}(6x^2-6x+1), \quad (70)$$

$$\hat{P}_3(x) = \sqrt{7}(20x^3-30x^2+12x-1). \quad (71)$$

4.3 Weighted Least Squares Polynomial Approximation

Assume that we want to approximate a function $f \in L_\mu^2(\Gamma)$, where $\Gamma \subset \mathbb{R}^d$ and μ is a probability measure on Γ . The strategy of weighted discrete least squares polynomial approximation is to

- choose a finite-dimensional space $V \subset L_\mu^2(\Gamma)$ of polynomials on Γ
- choose a function $\rho : \Gamma \rightarrow \mathbb{R}$ that satisfies $\int_\Gamma \rho(\mathbf{y}) \mu(d\mathbf{y}) = 1$ and $\rho > 0$
- generate $N > 0$ independent random samples from the *sampling distribution* ν defined by $\frac{d\nu}{d\mu} := \rho$,

$$\mathbf{y}_j \sim \nu, \quad j \in \{1, \dots, N\}. \quad (72)$$

Here, $\frac{d\nu}{d\mu}$ denotes the density, or Radon–Nikodym derivative, of the probability measure ν with respect to the reference measure μ .

- evaluate f at $\mathbf{y}_j$, $j \in \{1, \dots, N\}$
- define the *weight function* $w := \frac{1}{\rho} : \Gamma \rightarrow \mathbb{R}$
- and finally define the *weighted discrete least squares approximation*

$$\Pi_V f := \arg \min_{v \in V} \|f - v\|_N, \quad (73)$$

where

$$\|g\|_N^2 := \langle g, g \rangle_N := \frac{1}{N} \sum_{j=1}^N w(\mathbf{y}_j) |g(\mathbf{y}_j)|^2, \quad \forall g : \Gamma \rightarrow \mathbb{R}. \quad (74)$$

It is straightforward to show that the coefficients $\mathbf{v}$ of $\Pi_V f$ with respect to any basis $(B_j)_{j=1}^m$ of V are given by

$$\mathbf{G}\mathbf{v} = \mathbf{c}, \quad (75)$$

with $G_{ij} := \langle B_i, B_j \rangle_N$ and $c_j := \langle f, B_j \rangle_N$, $i, j \in \{1, \dots, m\}$, assuming that $\mathbf{G}$ is invertible. If $\mathbf{G}$ is not invertible, then (73) has multiple solutions and we define $\Pi_V f$ as the one with the minimal $L_\mu^2(\Gamma)$ norm.

Remark 2. We want to consider the L_μ^2 inner product $\langle g, h \rangle_\mu = \int_\Gamma g(y)h(y) d\mu(y)$ but we sample y_j from another distribution ν with density $\rho = \frac{d\nu}{d\mu}$ (w.r.t. μ). The method sets the weight $w(y) = \frac{1}{\rho(y)}$ and defines the discrete inner product

$$\langle g, h \rangle_N = \frac{1}{N} \sum_{j=1}^N w(y_j)g(y_j)h(y_j). \quad (76)$$

Take expectation over the draw $y \sim \nu$:

$$\mathbb{E}_\nu [w(y)g(y)h(y)] = \int_\Gamma w(y)g(y)h(y) d\nu(y) = \int_\Gamma \frac{1}{\rho(y)}g(y)h(y)\rho(y) d\mu(y) \quad (77)$$

$$= \int_\Gamma g(y)h(y) d\mu(y). \quad (78)$$

So the weighted average is an *unbiased Monte Carlo estimator* of the L_μ^2 inner product. By the law of large numbers,

$$\langle g, h \rangle_N \xrightarrow[N \rightarrow \infty]{\text{a.s.}} \langle g, h \rangle_\mu. \quad (79)$$

Let $\{B_i\}_{i=1}^m$ be an L_μ^2 -orthonormal basis of V . The LS normal matrix is

$$G_{ij} = \langle B_i, B_j \rangle_N. \quad (80)$$

Taking expectation and using the identity above,

$$\mathbb{E}[G_{ij}] = \langle B_i, B_j \rangle_\mu = \delta_{ij}. \quad (81)$$

Remark 3. Pick a basis $\{B_1, \dots, B_m\}$ of the polynomial space V and represent the unknown approximant as $v(y) = \sum_{i=1}^m a_i B_i(y)$ (coefficients $a \in \mathbb{R}^m$). Taking samples: $y_j \sim \nu$ with weights $w_j := w(y_j) = 1/\rho(y_j)$, the Weighted discrete inner product is:

$$\langle g, h \rangle_N = \frac{1}{N} \sum_{j=1}^N w_j g(y_j)h(y_j) \quad (82)$$

The Least-square objective is

$$J(a) = \frac{1}{N} \sum_{j=1}^N w_j \left(f(y_j) - \sum_{i=1}^m a_i B_i(y_j) \right)^2 = \langle f - v, f - v \rangle_N = \|f - v\|^2 \quad (83)$$

$$J(a) = \sum_{j=1}^N \left[\frac{1}{\sqrt{N}} \sqrt{w_j} f(y_j) - \frac{1}{\sqrt{N}} \sqrt{w_j} \sum_{i=1}^m a_i B_i(y_j) \right]^2 \quad (84)$$

$$= \sum_{j=1}^N \left[b_j - \sum_{i=1}^m M_{ji} a_i \right]^2 \quad (85)$$

$$= \sum_{j=1}^N [(b - Ma)_j]^2 \quad (86)$$

$$= \|b - Ma\|_2^2. \quad (87)$$

Equivalently, define the **design matrix**

$$M_{ji} = \frac{1}{\sqrt{N}} \sqrt{w_j} B_i(y_j), \quad (88)$$

$$b_j = \frac{1}{\sqrt{N}} \sqrt{w_j} f(y_j) \quad (89)$$

Then

$$J(a) = \|b - Ma\|_2^2 \quad (90)$$

Set the gradient to zero:

$$\nabla J(a) = 2M^\top(Ma - b) = 0 \implies M^\top M a = M^\top b. \quad (91)$$

Compute entries:

$$(M^\top M)_{ik} = \frac{1}{N} \sum_{j=1}^N w_j B_i(y_j) B_k(y_j) = \langle B_i, B_k \rangle_N =: G_{ik}, \quad (92)$$

$$(M^\top b)_i = \frac{1}{N} \sum_{j=1}^N w_j f(y_j) B_i(y_j) = \langle f, B_i \rangle_N =: c_i. \quad (93)$$

Hence the **weighted LS normal equations** are

$$Ga = c, \quad G_{ik} = \langle B_i, B_k \rangle_N, \quad c_i = \langle f, B_i \rangle_N. \quad (94)$$

4.4 Adaptive Algorithm

Here $\Gamma = [0, 1]^d$, $d \in \mathbb{N}$ and $\mu = \lambda$ is the Lebesgue measure.

Definition 1 (Multivariate Legendre polynomials).

- (i) We denote by $(P_i)_{i \in \mathbb{N}}$ the univariate $L_\lambda^2([0, 1])$ -orthonormal Legendre polynomials and define their tensor products

$$P_\eta := \bigotimes_{j=1}^d P_{\eta_j} : [0, 1]^d \rightarrow \mathbb{R}, \quad (95)$$

$$P_\eta(\mathbf{y}) := \prod_{j=1}^d P_{\eta_j}(y_j) \quad (96)$$

where $\eta = (\eta_1, \dots, \eta_d) \in \mathbb{N}^d$ is a multiindex indicating which polynomial degree is used in each coordinate.

(ii) For each multi-index $\mathbf{k} \in \mathbb{N}^d$, we define the polynomial subspace

$$\mathcal{P}_{\mathbf{k}} := \text{span} \{P_{\eta} : 2^{\mathbf{k}} - 1 \leq \eta < 2^{\mathbf{k}+1} - 1\} \subset L^2([0, 1]^d, \lambda). \quad (97)$$

In other words, $\mathcal{P}_{\mathbf{k}}$ is the span of those P_{η} whose multiindex components satisfy:

$$2^{k_j} - 1 \leq \eta_j < 2^{k_j+1} - 1 \quad \text{for each coordinate } j. \quad (98)$$

We use the notation $f_{-1} := 0$ to avoid separate treatment of the term corresponding to $l = 0$ in the following. To describe a multilevel approximation, we need to construct a sequence $(V_k)_{k=0}^L$ of polynomial subspaces, such that the difference $f_l - f_{l-1}$ is projected onto V_{L-l} using weighted least squares approximation. The final approximation is then defined as

$$\sum_{l=0}^L \Pi_{L-l}(f_l - f_{l-1}). \quad (5.1)$$

where Π_k projects onto V_k for $0 \leq k \leq L$. As in Section 4, if the samples used by Π_k are distributed according to the optimal distribution of V_k , then we require that the number of samples N_k satisfy

$$\kappa \frac{N_k}{\log N_k} \geq \dim V_k \quad (5.2)$$

for some $\kappa > 0$. As an alternative, we may use samples from the arcsine distribution, which is independent of the polynomial subspaces V_k . By Section 3.2, this increases the number of required samples only by a constant factor.

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